

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and
Commerce

Department of Information Technology (Bsc.IT Semester III)

Python for Data Science Practical

Practical - II

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Class :SYIT	Batch : 1
Date of Assignment :22/07/2026	Date/Time of Submission :22/07/2026

Questions:

2a. Write a python code to create nested tuples.

CODE:

```
Student = ("Rahul",51,("Bsc.IT","Semester III"),92)
print("Name = ",Student[0])
print("Roll No. = ",Student[1])
print("Department = ",Student[2][0])
print("Semester = ",Student[2][1])
print("Percentage = ",Student[-1])
print("Rahul_51")
```

OUTPUT:

```
===== RESTA
Name = Rahul
Roll No. = 51
Department = Bsc.IT
Semester = Semester III
Percentage = 92
Rahul_51
> |
```

2b. Write a python code to sort the nested tuple using sorted () function.

CODE:

```
# Sorted

data = ["Rahul", "Chandan", "Jay"]
sorted_data = sorted(data, key=lambda x: x[0])
print(sorted_data)
print("Rahul_51")
```

OUTPUT:

```
===== RESTART: C:\
['Rahul', 'Jay', 'Chandan']
Rahul_51
>
===== RESTART: C:\
['Chandan', 'Jay', 'Rahul']
Rahul_51
> |
```

2c. Write a python code to copy or clone list.

CODE:

```
list1 = ["Rahul", "Amit", "Vivek"]
list2 = list1.copy()

print("Original List:", list1)
print("Copied List:", list2)

print("Rahul_51")
```

OUTPUT:

```
===== RESTART: C:/Users/n  
Original List: ['Rahul', 'Amit', 'Vivek']  
Copied List: ['Rahul', 'Amit', 'Vivek']  
Rahul_51
```

2d. Write a python code to check immutability property of python tuples.

CODE:

```
my_tuple = ("Rahul","Raj","Nitish")  
my_tuple[1] = "Nitish"  
print(my_tuple)  
print("Rahul_51")
```

OUTPUT:

```
===== RESTART: C:/Users/mvlui/Desktop/Python/Python/PDS.py  
Traceback (most recent call last):  
  File "C:/Users/mvlui/Desktop/Python/Python/PDS.py", line 2, in <module>  
    my_tuple[1] = "Nitish"  
TypeError: 'tuple' object does not support item assignment
```

Curiosity Questions:

Q1. Define a tuple. How is it different from a list? Give one example.

Q2. Write a Python program to create the following tuples:

- a) An empty tuple
- b) A tuple with integers only
- c) A tuple with mixed data types
- d) A nested tuple (tuple inside another tuple)

Q3. Write a program to access elements of a tuple using indexing and slicing.

Q4. Create a tuple and find the following:

- Length of the tuple
- Maximum and minimum value (if all are numbers)
- Count of a specific item

Q5. Write a program to convert a list into a tuple.

Q6. Write a program to sort a tuple of numbers in ascending order.

Q7. Write a program to show that tuples are immutable (cannot be changed).

Q8. Write a program to clone (copy) a list and print both original and copied lists.

Q9. Write the output of the following code:

```
t = (1, 2, 3, 4, 5)
print(t[1:4])
```

Q10. Create a tuple of student marks: (78, 85, 92, 88, 76). Write a program to:

- Find average marks

- Display marks above 80